

Room Capture Protocol

Two-person field kit — 3BHK dimensional survey. Print this and carry it. Every number here is decided for you; do not improvise during the walk.

The four numbers that matter

Phone height	1.40 m	chest, camera level — never tilted up or down
Pass A standoff	1.00 m	one photo every 0.30 m along the wall
Pass B standoff	1.80 m	one photo every 0.50 m along the wall
Scale marker	1.000 m	two tape marks exactly 1 m apart, on the floor, in frame

YOUR TWO JOBS

PERSON A — CAMERA

never stops moving sideways

- Holds the phone **vertically**, 1.40 m, camera level, on the **1× lens**.
- Performs every photo pass on page 3, in the order given.
- Shoots **continuously while walking**. Do not stop to compose.
- Photographs the measurement card each time **B** reads a number out.
- Calls "**clear**" before shooting so **B** steps out of frame.

PERSON B — SURVEY & MEASURE

never in the photo

- **Before shooting**: walk the room, count corners, doors, windows, mirrors and false ceiling. Call the list aloud so **A** knows the target count.
- Lay the **1 m scale marker** on the floor at the room start.
- Stick **masking-tape marks** on the floor at every corner. These anchor the tape later.
- **Park out of frame** during shooting, or leave the room entirely.
- **After shooting**: measure with the 5 m tape, writing every number on the card.
- **Tick the acceptance checklist** (page 6) before leaving the room.
- Write the **room name** on the card before measuring. Never measure an unnamed room.

Why you cannot do both jobs at once

Shooting and measuring interfere: a tape left in frame pollutes the photo set, and a measuring posture breaks the walking rhythm the photo passes depend on. So the room cycle is always **shoot first, measure second**. Never interleave them.

HOW MANY ROOMS

Photographing is **fast** — about 12 minutes for a mid-size room. Measuring is **slow** — about 28 minutes. So capture **every** room you can reach, but only measure the top three. A room photographed but not measured still helps; a room measured but not photographed helps nothing at all.

TIER	ROOMS	PHOTOS	MEASURE?	TIME
1 — full	Living room, Master bedroom	~167 ea	yes, full	55 min
2 — standard	Kitchen, Bedroom 2	~110 ea	yes, full	40 min
3 — quick	Dining, Bedroom 3, passage	~55 ea	no	8 min ea
skip	Bathrooms, balcony, utility	—	no	0
Whole flat, realistic total		~600	3 rooms	2.5–3 h

Skip the bathrooms: small, cluttered, mirror-heavy, and the lowest value per minute in the flat. Revisit only if you are well ahead of schedule.

The room cycle

Steps 1–5 for every room. Never skip step 1 — it is what stops you photographing a room with the fan spinning and a door half shut.

Step 1 — Make the room still **B leads · 30 s**

- ☐ Ceiling fan **OFF**. Pedestal fan **OFF**. Air conditioning off.
- ☐ Remove or push down loose cushions, clothes on chairs, movable floor objects.
- ☐ Curtains **fully open or fully closed — then do not touch them** for this room.
- ☐ Doors and windows **open**. You need the reveal and the true opening width.
- ☐ Switch on every light in this room now and leave them on for the whole session.
- ☐ No person, no pet, no tripod anywhere in frame.

Step 2 — Survey and count **B leads · 60 s**

Walk the perimeter once, calling out and counting:

- ☐ Number of **corners** — write the number on the card
- ☐ Every **door** and every **window**, with sill heights
- ☐ Every **mirror** — flag each one; they corrupt depth silently
- ☐ **False / POP ceiling?** If yes, note it — ceiling height becomes ambiguous
- ☐ Any **damage**: cracks, chips, missing tiles, stains, water marks

These counts are A's shot targets. Without them A does not know when to stop.

Step 3 — Lay the anchors **B does · 90 s**

- ☐ Lay **1.000 m of tape flat on the floor** with a masking-tape mark at each end, near a corner but inside the room. This is the scale reference — do not skip it.
- ☐ Stick a small masking-tape cross on the floor at **each corner**. These are the tape's anchor points and the photo-to-measurement correspondence.

Step 4 — Photograph **A does · ~12 min**

Run the eight passes on the next page, in order. **B** leaves the room or parks in a corner well out of frame. **A** starts at the scale marker and walks **anticlockwise**, one full loop per pass, so every pass covers the same walls in the same order — that is what makes gaps obvious.

Step 5 — Measure **B leads · ~28 min**

Tier 1 and Tier 2 rooms only — full list on page 5. **A** photographs the card with each number legible, filling the frame.

If you have only 90 minutes, cut in this order

1. Living room: passes 1, 2, corners, openings, scale marker. Then measure it.
2. Master bedroom: same. Then measure it.
3. Stop there. Everything else is optional — two rooms done properly beats seven done badly.

Never photograph a room you will not measure, and never measure a room you did not photograph.

The eight passes — angles and counts

Per room. Counts assume a 4.5 × 3.6 m room (16 m perimeter, 4 corners, 2 doors, 2 windows, 5 features). Scale the shot counts by your wall length.

#	PASS	HOW	SHOTS	TIME
1	Perimeter A the essential one	1.00 m off the wall, phone at 1.40 m, moving sideways only , facing the wall, one photo every 0.30 m (about one slow step). Full loop.	~55	3 min
2	Perimeter B	From the opposite wall, ~1.80 m out, same rhythm, one photo every 0.50 m . A second ring breaks the degenerate single-loop path — this is what lets scale separate cleanly from the two wall directions.	~33	2 min
3	Corners	Stand <i>inside</i> each corner. Three photos: one at each wall, one straight at the junction. The most important 12 shots in the room — the wall-wall junction fixes both the room polygon and the angle between walls.	12	1 min
4	Junction tilts	At each corner, tilt down for the floor-to-wall seam and up for the ceiling-to-wall seam. A level photo of a wall never shows where it meets the floor.	8	1 min
5	Floor pass	Camera at ~1.7 m, tilted ~45° down, walk across the room with overlap. Skip only if the floor is blank polished stone with no joints.	6	0.5 min
6	Ceiling pass	Tilt ~45° up, small overlapping loop. Only if there is no false ceiling . With a POP ceiling, photograph both levels instead — see page 4.	6	0.5 min
7	Openings highest scoring	Each door and window, square-on from 1.5 m, 5–6 photos stepping left to right at 0.30 m spacing . Include the full height of the opening plus surrounding wall. Openings are the highest-accuracy gate in the entire assessment.	~24	2 min
8	Features & damage	3–5 overlapping close shots at 0.5–0.8 m of each switchboard, socket, skirting, cornice, crack, chip and stain. The surface should fill the frame and be in focus.	~20	2 min
Tier 1 total		Plus 3 photos of the 1 m scale marker	~167	12 min

The three mistakes that cost the most

- 1. Panning in place.** Standing in the middle and rotating the camera produces almost no geometric information — rotate-only motion is nearly degenerate. You must **translate sideways**. Passes 1 and 2 are the only passes that carry real geometry.
- 2. Stopping to frame each shot.** Every stop-and-compose breaks the overlap chain and wastes the parallax of continuous motion. Walk steadily, shoot roughly every step, review afterwards.
- 3. Skipping the corners.** Every shortcut gets taken at the corners, and corners are exactly what determine a room's shape. Budget the full 12 shots.

Settings, lighting, hazards

PHONE SETTINGS — SET ONCE, THEN LEAVE IT ALONE

SETTING	DO THIS	WHY
Lens	1× only. Never 0.6× ultra-wide, never 2×/3× zoom	Ultra-wide has heavy barrel distortion; digital zoom crops and resamples, destroying texture
Lens glass	Clean it before you start	Fingerprints cost more accuracy than any other setting on this page
Resolution	Full resolution, JPEG	Highest usable texture
Exposure	Lock it, if the phone allows (Pro / Expert mode)	Auto-exposure pumping between frames is a genuine failure mode on dim walls
White balance	Lock it, if possible	Keeps colour true for material and damage classification
Flash	OFF, always	Flash off a wall at 1 m makes a hotspot with hard falloff — both kill texture
Horizon guide	Turn on if the phone has one	Passes 1–2 require a level camera
Storage & battery	5 GB+ free, battery 100%	~600 photos is roughly 4.5 GB
Orientation lock	On, portrait	Prevents a mid-session rotation

Lighting — pick your moment, do not fix it later

Best: overcast, or the hour after sunrise, or the last hour before sunset. **Worst:** bright sun through a west-facing window, which in an Andhra summer produces hard shadow edges and blown highlights no setting recovers.

- Never have a window **behind** the wall you are shooting — the wall goes to silhouette.
- Never let a person's or a fan's shadow fall across a wall you are shooting.
- Turn **off** any flickering LED, any TV, and every moving fan before you begin.
- If a room is dim, **switch on its existing lights and stay**. Do not chase better light — you will lose the schedule.

Four hazards that silently corrupt results

These do far more damage than any camera setting. Note each one's location: a *named* limitation is acceptable, a *silent* wrong number is not.

1. **Mirrors.** Monocular depth models return a plausible, confident, **wrong** depth on a mirror instead of failing. Note every mirror. Never trust a measurement taken near one.
2. **Glass and balcony doors.** Same failure, plus see-through geometry.
3. **POP / false ceiling.** A dropped ceiling below the slab makes "ceiling height" ambiguous by 10–20 cm. Photograph both levels, and **state which one you measured**.
4. **Glossy surfaces** — polished granite floor, glossy modular kitchen. Specular reflection leaves no usable texture. Photograph edges and matte areas only.

Painted featureless walls are the hardest common case — no texture to match on. Get a light switch, curtain rail, ladder or cornice into shots near bare wall; those are your texture anchors.

Ground truth — measuring with the 5 m tape

Tier 1 and Tier 2 rooms only. These numbers are the truth every other number is checked against, so precision matters more than completeness.

One rule, and it matters. The tape is your **validation** reference. The pipeline gets its scale from the **1 m marker photographed in the scene**. Never calibrate the pipeline against these hand-measured numbers — that makes the benchmark circular and worthless, and a reviewer checks for exactly this.

MEASURE THIS LIST — ABOUT 25 PER ROOM

WHAT	COUNT	HOW
Wall lengths, corner to corner at floor level	4	Between the masking-tape corner marks. Two per axis.
Wall heights, floor to ceiling	4	One per wall, at mid-wall, tape held vertical.
Floor diagonal	1	Corner to opposite corner. Checks whether the room is really rectangular.
Opening widths	3 each	Top, middle and bottom — openings are often trapezoidal, and that variation is real information.
Ceiling height	3	Not five. Different parts of the room, and under any false ceiling.
Window sill height	1 each	Floor to the underside of the sill.
Floor landmark positions	2–3	The tape marks already on the floor. Note which corner each sits near.
Total	~25	~28 minutes with two people

Solo measuring technique

- **Use the masking-tape corner marks as anchors.** Hook the tape's metal end on a mark, pull to the next mark, read, write, step. You never need a second hand at the far end.
- **For runs over 5 m,** step the tape in marked segments and sum them. Mark every intermediate point so a later re-measure is reproducible.
- **Pull the tape taut and straight.** A sagging tape reads short; a twisted one reads long. Sight along it.
- **Read at eye level,** directly above the tape, never at an angle.
- Record **millimetres**, not centimetres, with leading zeros: 1245, not 1 245.
- **Measure every wall length twice,** independently, and write both numbers. If they differ by more than 5 mm, measure a third time and keep all three.

Measurement card — one per room

A single index card in large handwriting. **A** photographs it after every five readings, filling the frame, so each number stays permanently linked to its room.

Room name (written before measuring)	LIVING ROOM
Date	04 Oct
Wall lengths L1 / L2 / W1 / W2	4250, 4180 / 3620, 3640 mm
Wall heights H1–H4	2740, 2755, 2730, 2748 mm
Diagonal D1	5582 mm
Door widths (top/mid/bot)	905 / 900 / 902 mm
Window widths (top/mid/bot)	1450 / 1455 / 1448 mm
Ceiling heights, 3 spots	2740, 2755, 2730 mm
Sill heights	900 mm
False ceiling? which height measured?	YES — POP at 2400, slab not visible
Mirrors and glass, located	1 mirror dining wall; 1 glass door balcony

Acceptance checklist — tick before leaving each room

B owns this page. Do not move to the next room until every line is ticked or explicitly given up. A room photographed badly is a lost room — you will not come back for it.

SCENE

- ☐ Fan off, AC off, no moving person, no pet
- ☐ Doors and windows open
- ☐ Curtains fully open *or* fully closed, and untouched since
- ☐ All room lights on

COVERAGE

- ☐ Pass 1 complete — full loop at 1.0 m, spacing 0.30 m
- ☐ Pass 2 complete — second loop from the opposite wall at 1.8 m
- ☐ All **four corners** shot — 3 photos each, standing inside the corner
- ☐ Floor-to-wall and ceiling-to-wall tilts at each corner
- ☐ Floor pass done
- ☐ Ceiling pass done, **or** POP ceiling photographed at both levels
- ☐ Every door photographed — 5–6 photos each, square-on, full height
- ☐ Every window photographed — 5–6 photos each
- ☐ Every feature and every damage item photographed at 0.5–0.8 m

ANCHORS — THE TWO THINGS THAT CANNOT BE REDONE LATER

- ☐ **1.000 m scale marker** on the floor, photographed in 3 frames with the tape legible
- ☐ Masking-tape corner marks placed and their positions noted

GROUND TRUTH — TIER 1 AND TIER 2 ONLY

- ☐ Room name on the card **before** measuring
- ☐ All ~25 measurements written in millimetres
- ☐ Every wall length measured twice, both numbers written
- ☐ Opening widths taken at top, middle *and* bottom
- ☐ False ceiling noted, and which height was measured stated explicitly
- ☐ Every mirror and glass surface located and noted
- ☐ Card photographed filling the frame

TECHNICAL

- ☐ Zero photos taken on 0.6× ultra-wide or 2× zoom
- ☐ EXIF preserved — no screenshots, no re-saving, nothing sent through WhatsApp or Instagram
- ☐ First photo of the room checked for correct orientation
- ☐ Photo count matches the expected pass count — 0 means a broken loop, 300 means composing instead of sweeping

Wrap-up

Troubleshooting for when the session goes sideways, and the checks to run before you leave the building.

If something goes wrong

- **Fan will not stop, curtain will not stay** — photograph anyway, note it, move on. Do not lose the schedule.
- **Room is far too dark** — turn on its lights and continue. Never use flash.
- **You are running late** — drop passes 5 and 6 (floor and ceiling) first, then shorten pass 2 to a half loop. **Never drop corners or openings.**
- **A wall is blank painted plaster** — get the switchboard, cornice, skirting or door frame into shots near it. Those anchors are what make it reconstructable.
- **You find damage worth documenting well** — spend the extra 2 minutes. Take 5 overlapping close shots plus one wide shot for context, with a tape in frame for scale.

Before you leave the flat — final 10-minute check

- ☐ Photo count is roughly what you expected (~600 for the full flat)
- ☐ Every measured room has its card photographed with the room name legible
- ☐ The 1 m scale marker was photographed in at least two rooms
- ☐ Photos were **not** sent through WhatsApp, Instagram or any compressor
- ☐ Everything backed up to a second location before you leave
- ☐ You roughly know how many photos each room has, in case a set needs re-shooting

Hard cutoff

If you reach **13:00** and the Tier 1 rooms are not finished, stop photographing new rooms and start measuring what you already have. A measured room you photographed beats an unmeasured room you did not. Everything below that line can be recorded as a known gap — and an honest gap costs far less than a broken pipeline.